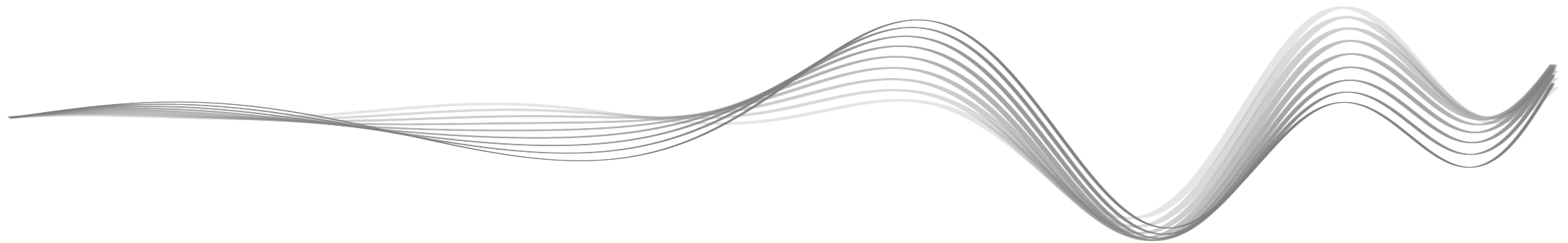




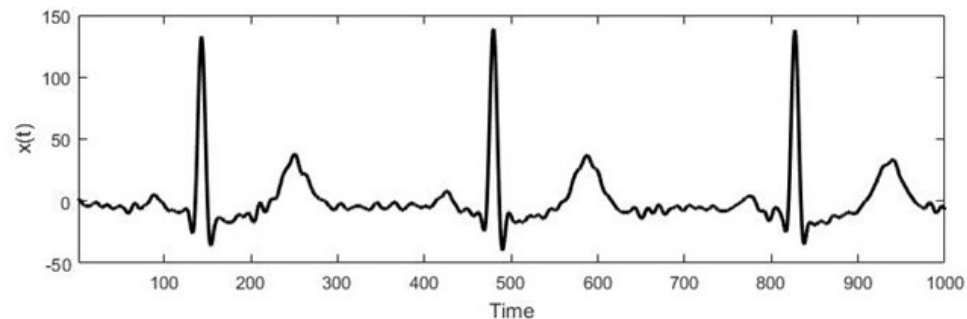
Deep Learning for Time Series





Time Series (TS)

Some Examples

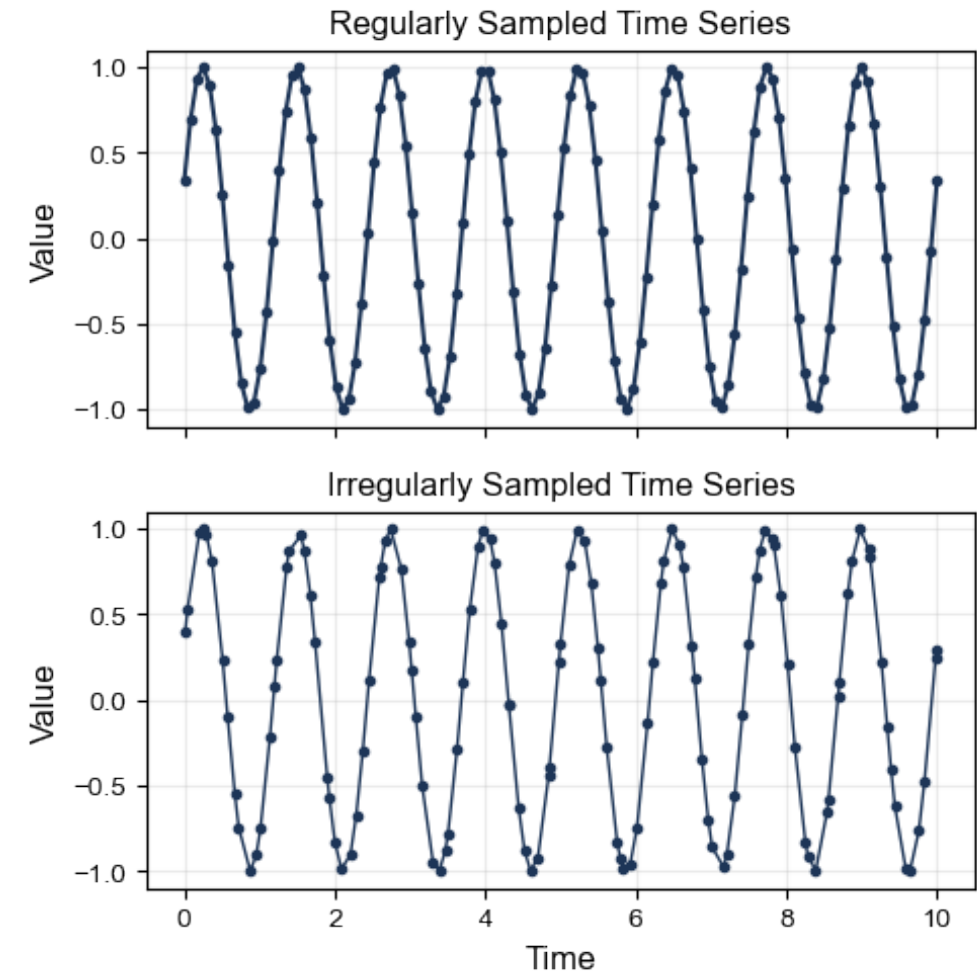


P. Čepulionis and K. Lukoševičiūtė, “Electrocardiogram time series forecasting and optimization using ant colony optimization algorithm,” *Mathematical Models in Engineering*, Vol. 2, No. 1, pp. 69–77, Jun. 2016.

What is a Time Series?



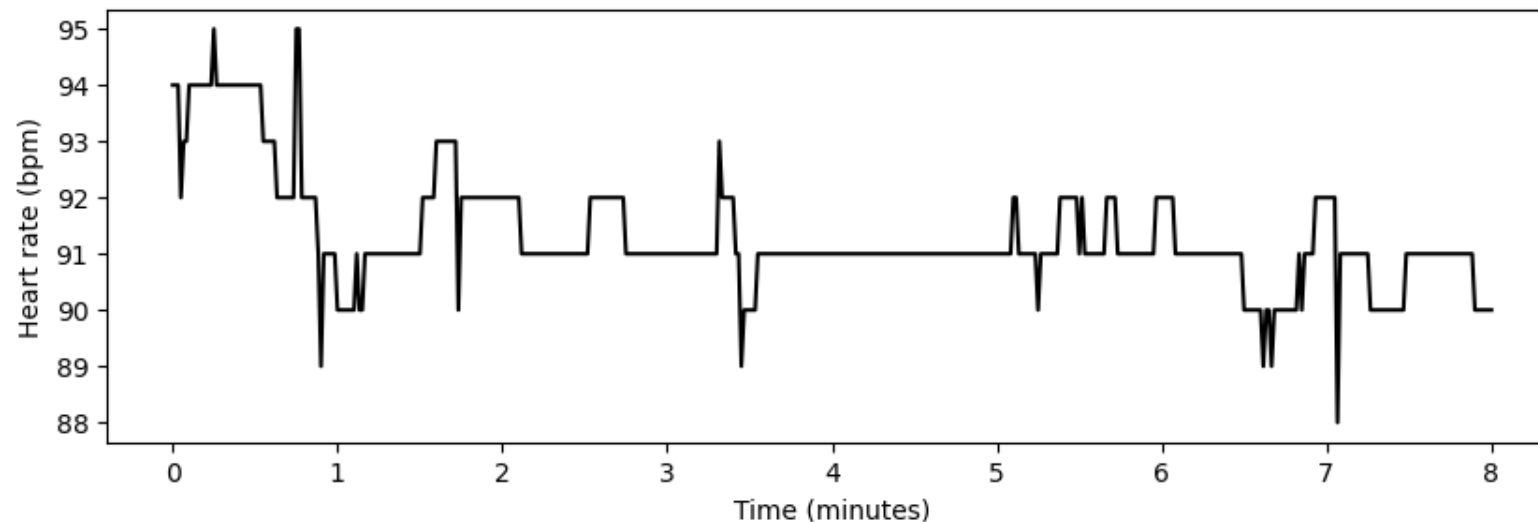
- **Time series:** a sequence of observations recorded at successive points in time
 - Real world: continuous data
 - Digital world: discrete-time data
- **Regular sampling:** constant time increments $\Delta\tau$ between measurements. *Considered hereafter*
- **Irregular sampling:** measurement at non-uniform or unequal time intervals
 - Event-driven measurements: patient visits doctor; stock prices recorded as individual trades happen; ...
 - Sensor faults: shut-off; network connection missing; ...



Uni- and Multivariate Sequences

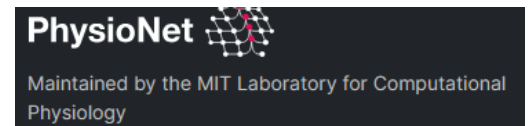


- **Univariate time series:** single-channel measurement,
e.g. basic fitness tracker records only the patient's heart rate every minute



PhysioNet, bidmc dataset

first patient's data



- Mathematical notation: $\mathbf{x} = [x_1, x_2, \dots, x_T]^T \in \mathbb{R}^{T \times 1}$
- Physical time: $\tau_t = \tau_0 + t\Delta\tau$

T : number of time steps

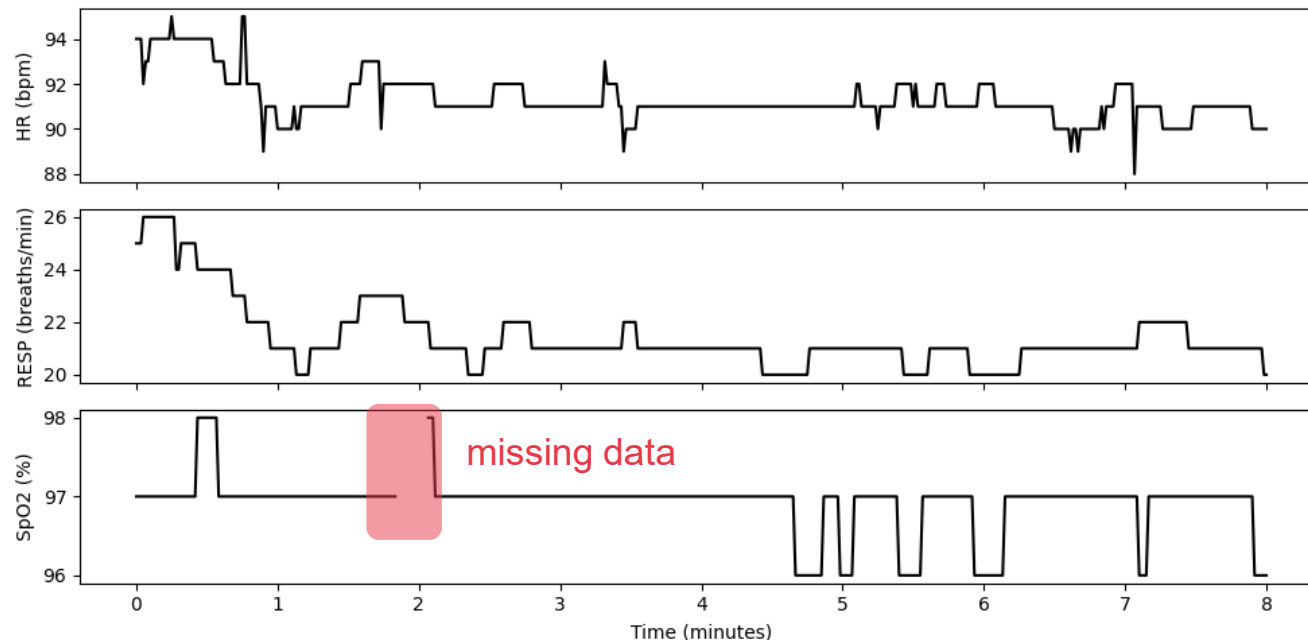
t : index of time step $t = 1, \dots, T$

Uni- and Multivariate Sequences



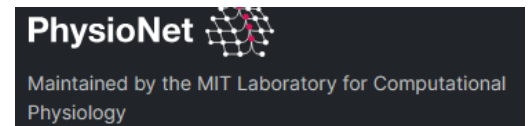
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- **Multivariate time series:** multi-channel measurement, e.g. advanced instrument recording heart rate, respiratory rate and oxygen saturation every minute



PhysioNet, bidmc dataset

first patient's data



- Mathematical notation: $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T]^T \in \mathbb{R}^{T \times C}$
 $x_{t,c}$ value of channel c at time step t

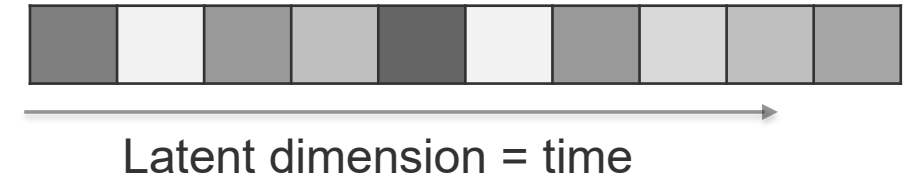
T : number of time steps
 t : index of time step $t = 1, \dots, T$

C : number of channels
 c : index of time steps $c = 1, \dots, C$

What makes TS so special?



- **Order matters:** $[x_1, x_2, x_3] \neq [x_2, x_3, x_1]$
 - Shuffling time steps destroys information
 - **Order of measurements is not interchangeable!**
- **The future must not leak into the past**
 - a model may only use information that would genuinely have been available at that moment
 - Data splitting requires care
- **Statistical behavior changes over time**
 - Trends, seasonality, transients
- Relevant information exists at **different time scales**
 - Relevant information stored in last sec/min/... year



Examples:

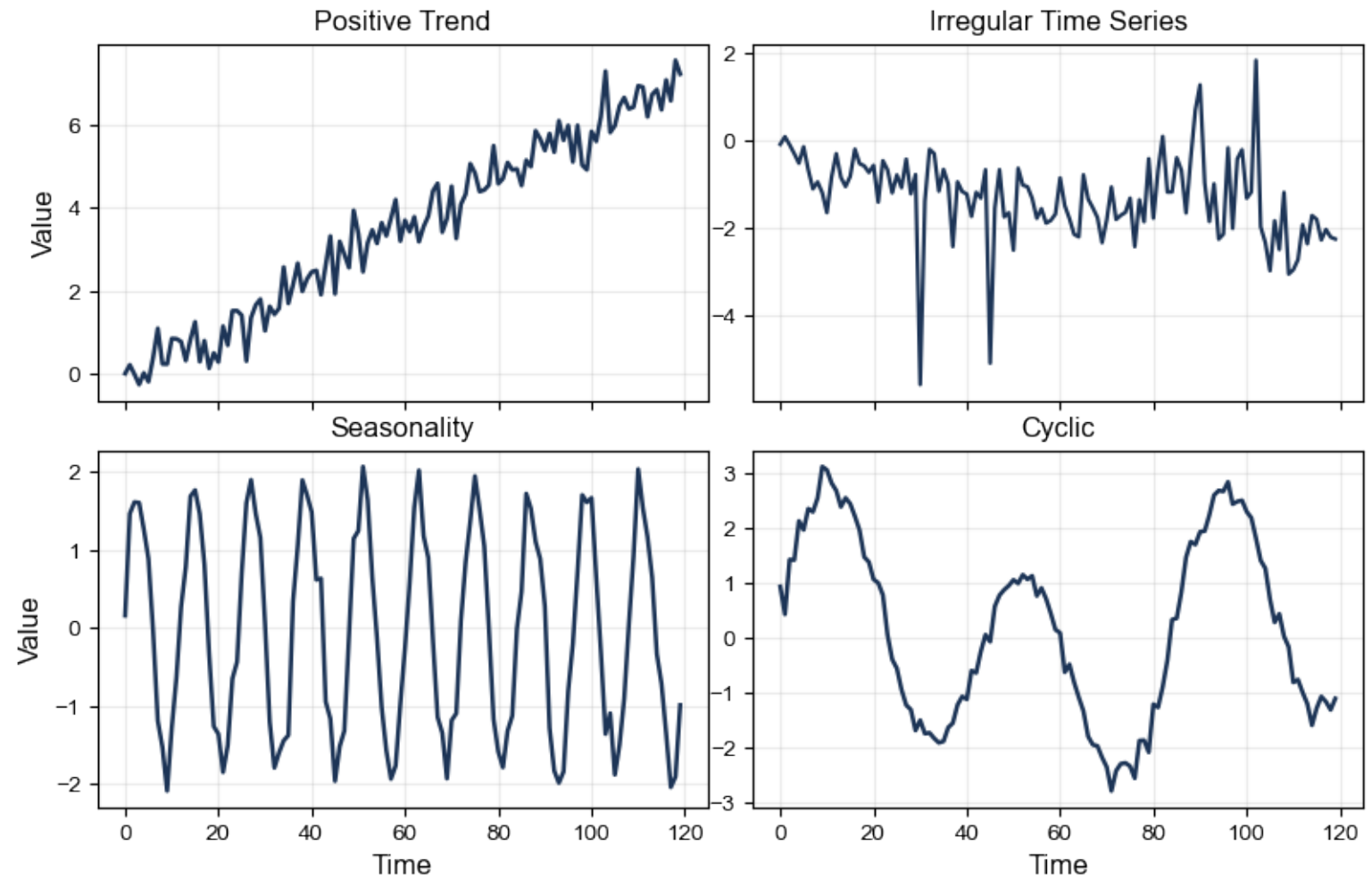
[I | live | in | Munich | but | work | in | Berlin]



Characteristics



- Seasonality
- Cyclicity
- Trend
- (Temporal) Irregularity



Seasonality



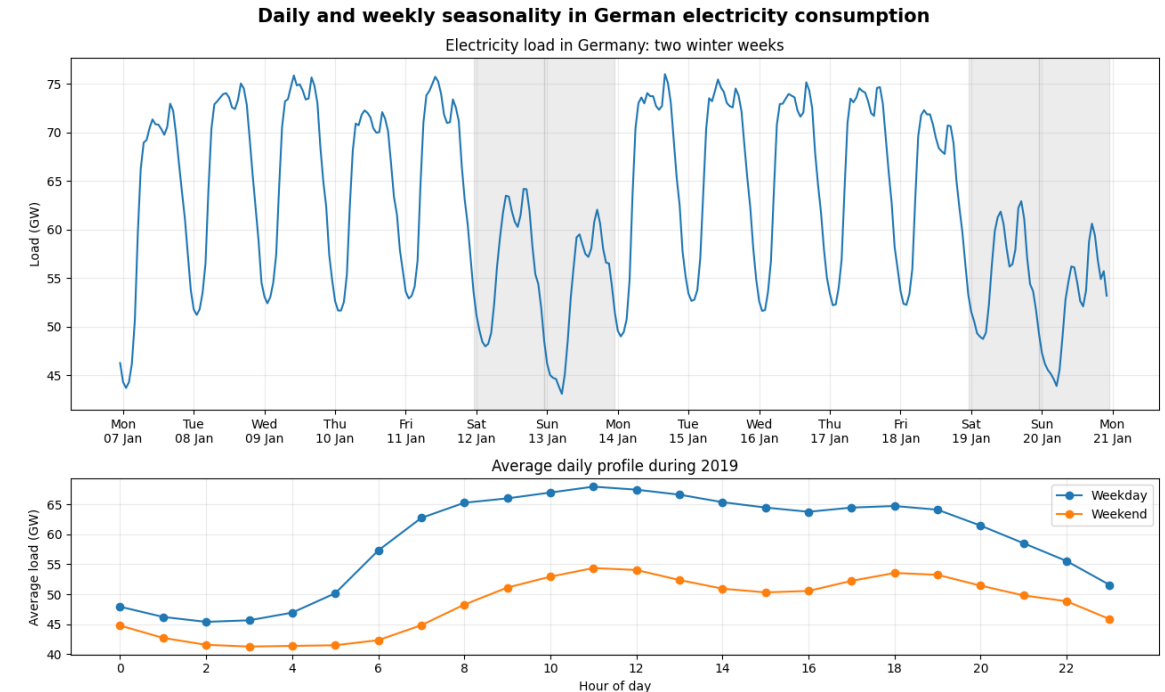
- A **pattern repeats at a known, regular interval S**

$$x_t \approx x_{t-S}$$

- not limited to annual cycles
- several seasonalities can occur simultaneously (single vs. multiple seasonality)

- Examples with seasonal period

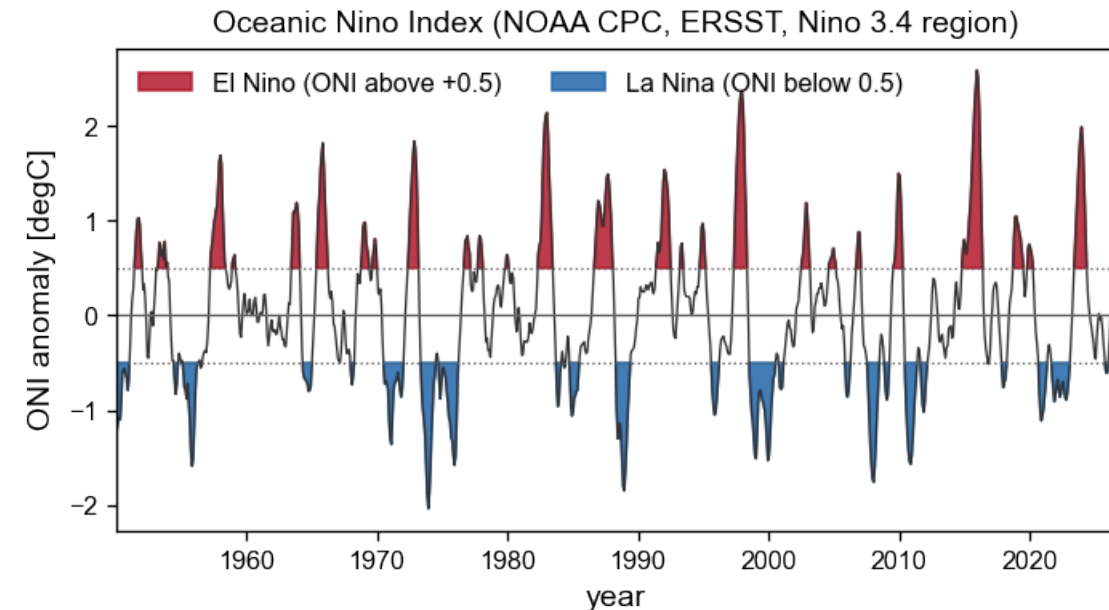
- Household power consumption (24 hours)
- Power demand (city, country) (7 days)
- Restaurant customers (7 days + 24 hours)
- Ocean height (12.4 hours)
- ...



Cyclicity



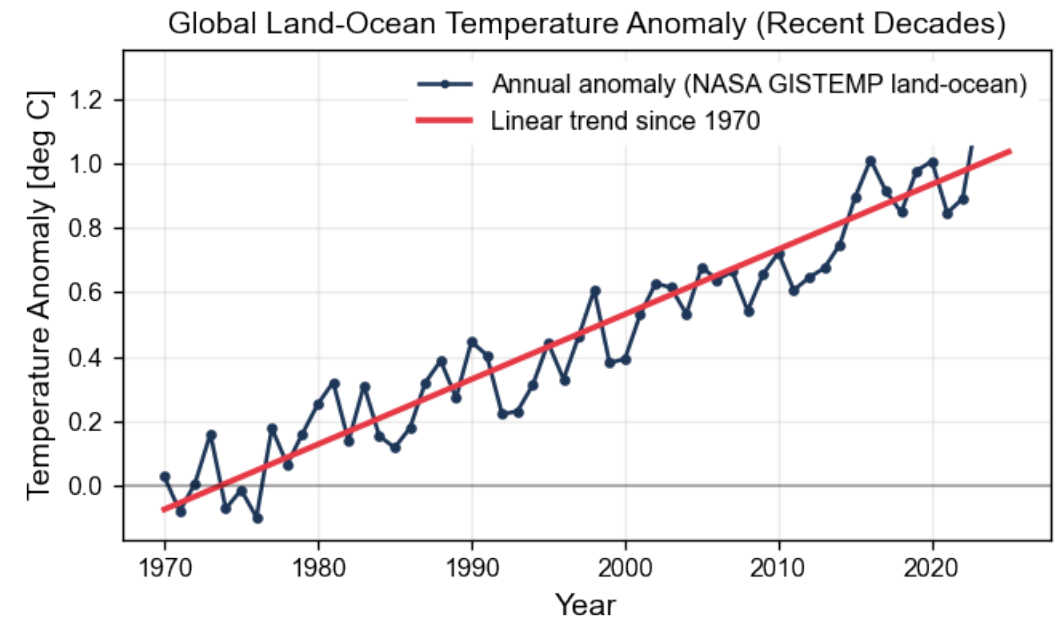
- A pattern recurs at an unknown or variable interval. $x_t \approx x_{t-s}$
- Examples:
 - Economic recession cycles
 - Climate phenomena (El Nino)
 - Machine operating cycles



Trend



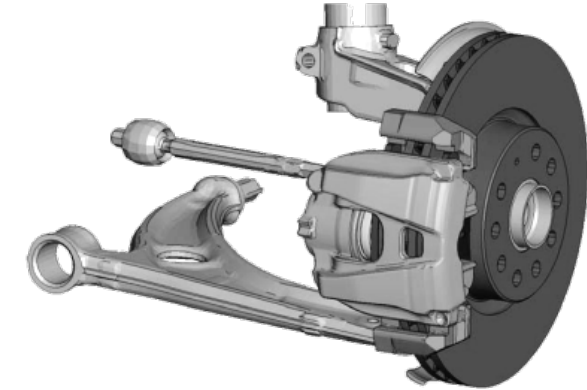
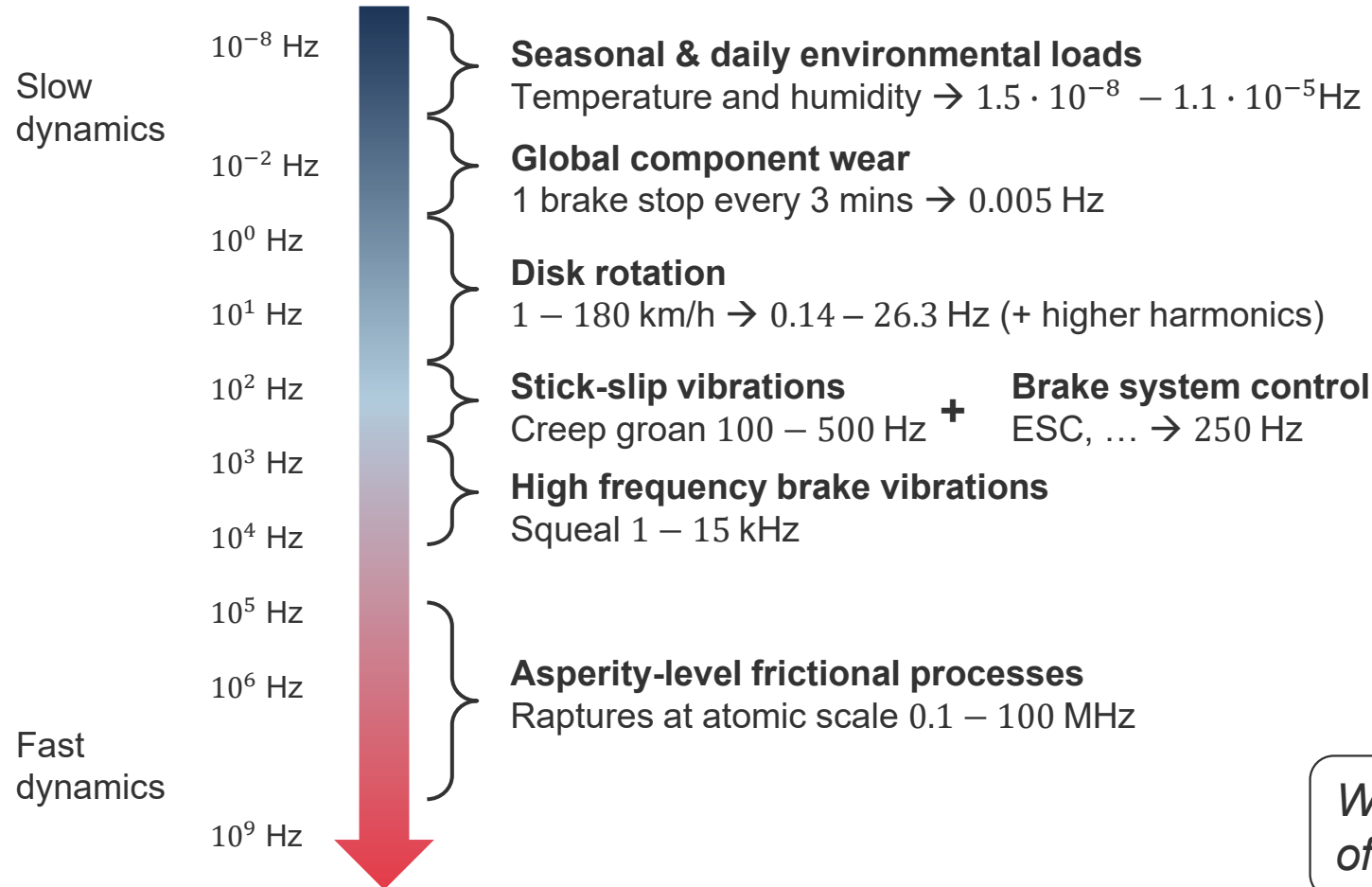
- **Trend:** gradual tendency of average series values to increase or decrease over time
 - Local trend: limited tendency over some time range
 - Global trend: tendency over complete measurement time range
- **Different forms of trends:**
 - Linear
 - Polynomial
 - Exponential
 - ...
- **Examples:**
 - Global ocean temperatures (global warming)
 - Stock market indices
 - ...





- **Irregular dynamics:** temporal variations that have neither a persistent direction nor a reliably repeating pattern
- Examples:
 - Unpredictable fluctuations (turbulence)
 - Abrupt events or transients (financial returns)
 - Changing amplitudes and frequencies (weather variability)
 - Switching between operating regimes (machine faults)
 - Chaotic or stochastic behaviour (human activity)
- Irregular $\neq$ random!
 - **Stochastic irregularity:** generated partly by noise or random influences.
 - **Chaotic irregularity:** generated by a deterministic system with high sensitivity to its initial state.
 - **Transient irregularity:** caused by isolated events, faults, or regime changes.

Multiple scales



- **Multi-scale dynamics**
- **Strong coupling between scales**
- **Non-stationary behavior**

Which time scales matter, how much of time history to take into account?



Deep Learning for TS

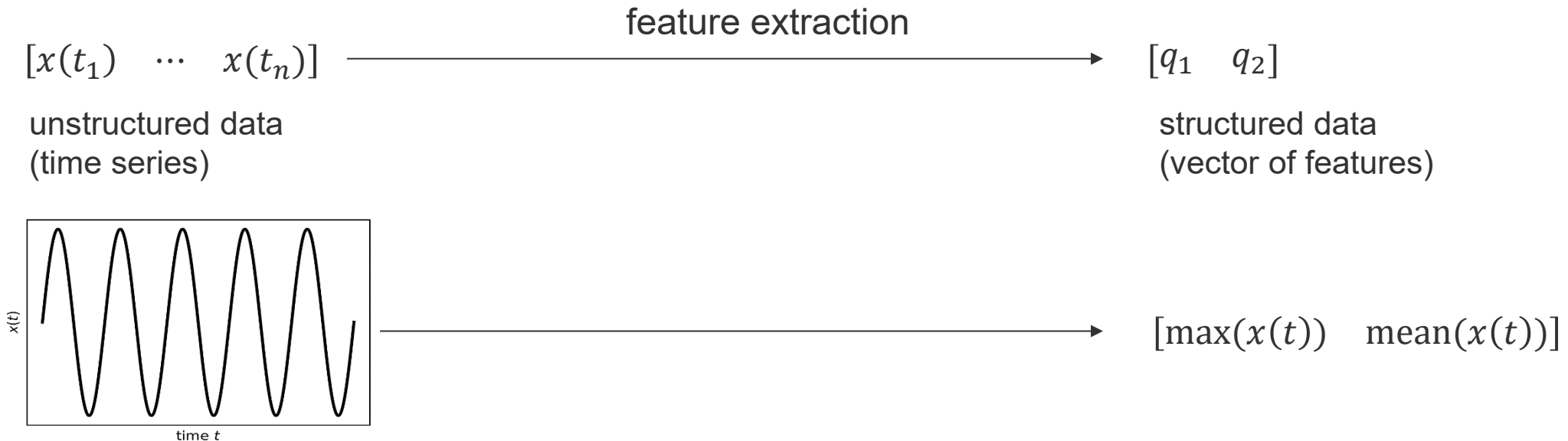


- **Time series classification:** recording of a bird's song → map to bird's species
- **Time series clustering** (unsupervised / self-supervised): anomaly detection
- **Time series regression:**
 - Sequence-to-sequence: virtual/soft sensor
 - Sequence-to-vector: remaining useful lifetime prediction, parameter identification
- **Time series forecasting**
 - Single-step
 - Multi-step
- **Approaches:**
 - Time-series only: models consume time series directly (or representations of it).
 - Featurization: tabular data of features is extracted from time series first

Feature Engineering



- **Machine learning:** structured data
- **Deep learning:** structured + unstructured data
- How to process unstructured engineering data (here: time series data) with ML?

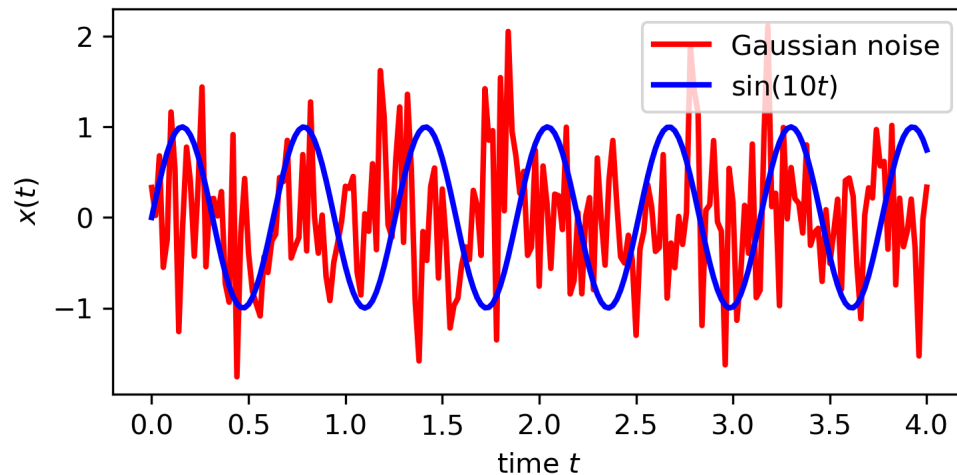


- Requires domain expertise!

Example: Signal Classification



- Classify vibration data: a) **harmonic signal** and b) **random (Gaussian) process**
- Obvious to the human eye – how to teach the computer?
- **Without** feature engineering → deep learning required
- **With** feature engineering → simple decision tree



Statistical characterization

Not generic (amplitude / frequency variation?)

```
maximum of noise signal: 1.726001130529702
maximum of sine signal:  0.9999118601072672
mean of noise signal:    -1.7675192431843288e-17
mean of sine signal:     0.04318132362329466
median of noise signal:  0.017201635395270597
median of sine signal:   0.10071709699250761
standard dev. of noise signal: 0.7093145682637918
standard dev. of sine signal: 0.7093145682637918
```

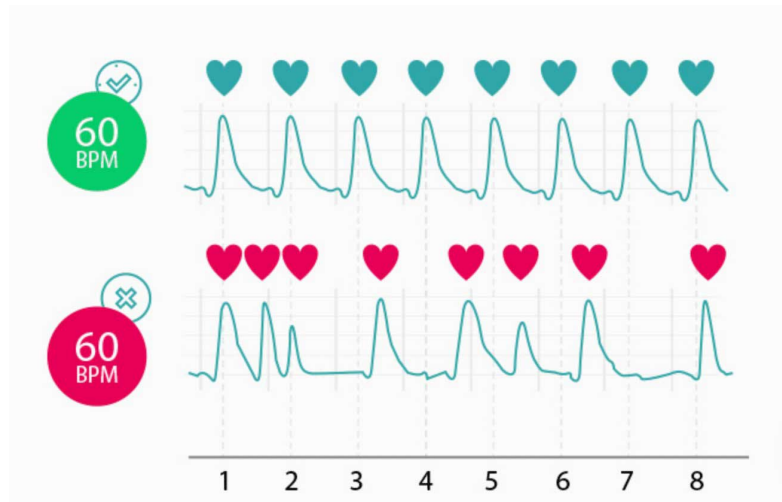
Better: signal processing methods
FFT, Nonlinear Time Series Analysis

Time Series Classification



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- Categorize time series

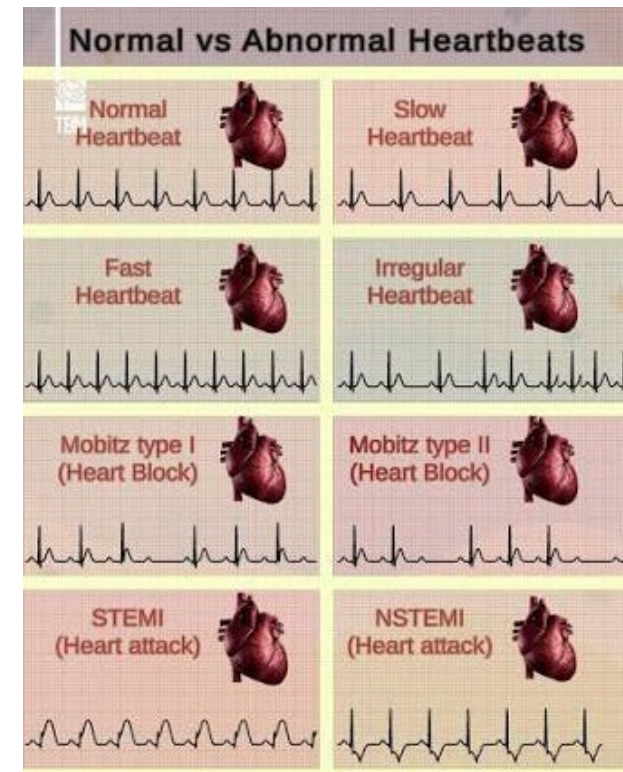


Cardiac anomalies (unsupervised)

<https://www.fibrichck.com/heart-rate-versus-heart-rhythm/>

Multi-class classification (supervised)

© thebrainmaze





5min break



Applications DL4TS

(from our research)

Marine Learning and Decision Support



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Ship-helicopter operation limits



Ship-to-structure transfer

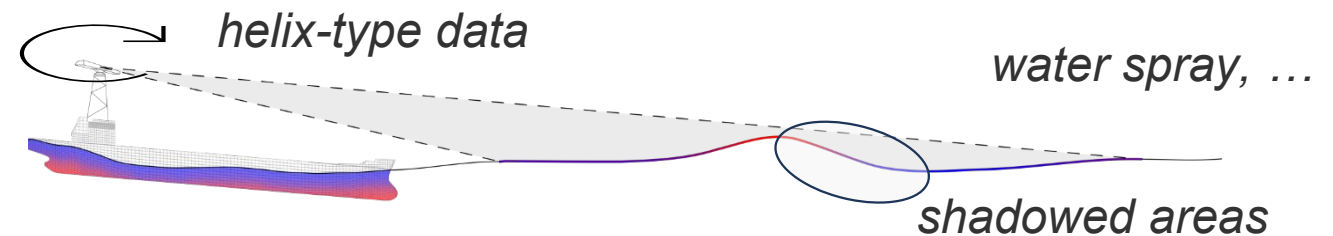


Marine construction



Short term (<5min) **wave forecasts** $\eta(x, y, t)$ required

- Radar: initial conditions $\eta(x, y, t_0)$
noisy, corrupted, sparse
- Numerics: forecasting
too slow



© Marco Klein, DLR

Ocean Wave Forecasting



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- 1D waves $\mathbb{R}^{(8 \times 1024)} \mapsto \mathbb{R}^{1024}$

predicting >15 peak periods

auto-regressive set up

4000m spatial domain

- 2D waves

high-quality
simulations



Models:

- U-Net (fully convolutional)
- 85k parameters (1D)
- 18k parameters (2D)

Data:

- 57 GB (1D)
- 3.4 TB (2D)

Training times

- ~30 hours (1D)*
- >1 week (2D)**

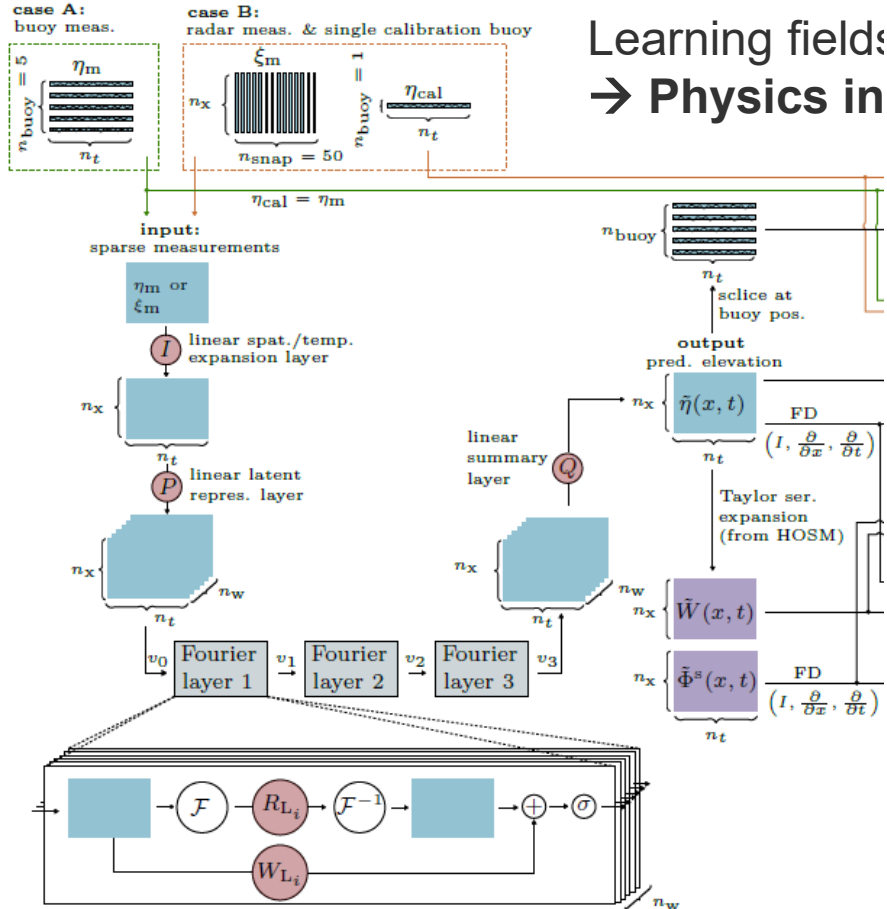
* Titan RTX3090
24GB VRAM

** Dual Nvidia A100
160GB VRAM

Learning Fields From Noisy Point Clouds



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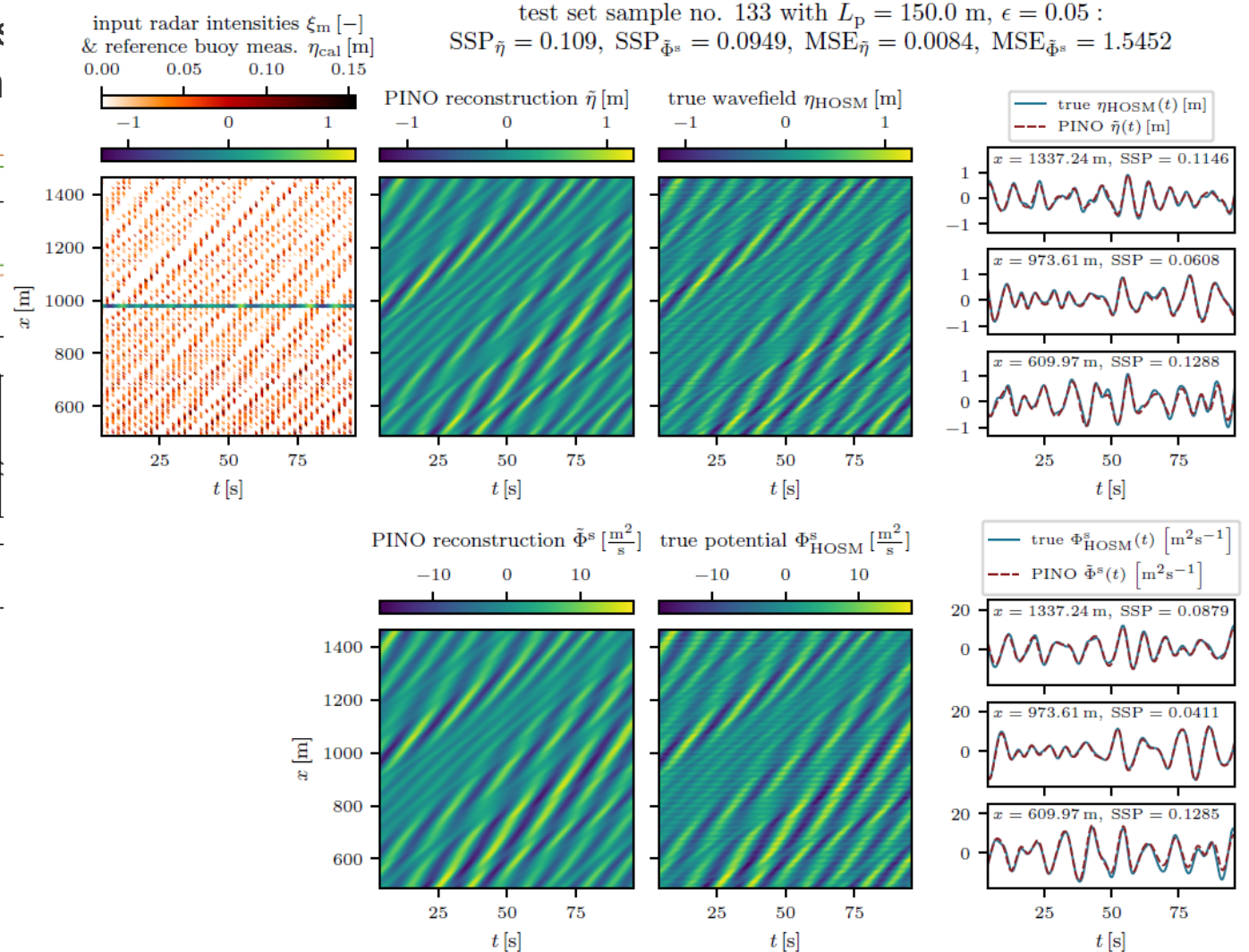


Ehlers, Stender et al. (2023) *Ocean Engineering*

Ehlers, Stender et al. (2024) *Fluids*

Ehlers, Stender et al. (2025) *Physics of Fluids*

Physics + Data + AI



Differentiable Physics Simulations



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Core idea:

- Keeping a **physics-based mathematical structure**
- Adding **learnable terms / parts / modules**
- **End-to-end differentiation**

Enables

- **Scalability** through vectorization
- **Explicit uncertainty** analysis (Monte-Carlo)
- **Learning unknown relationships** from **data***

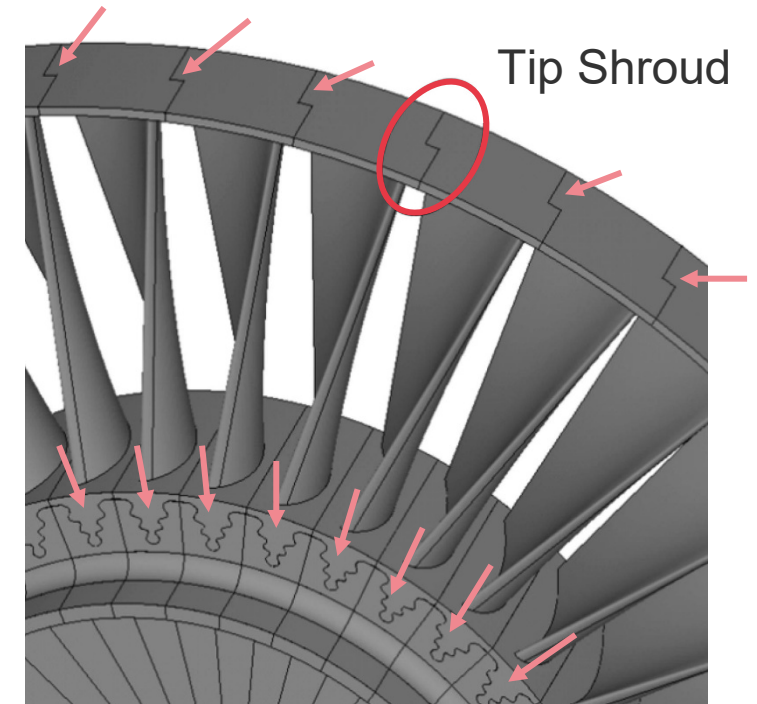


Image source: [<https://doi.org/10.1016/j.ymssp.2022.110080>]

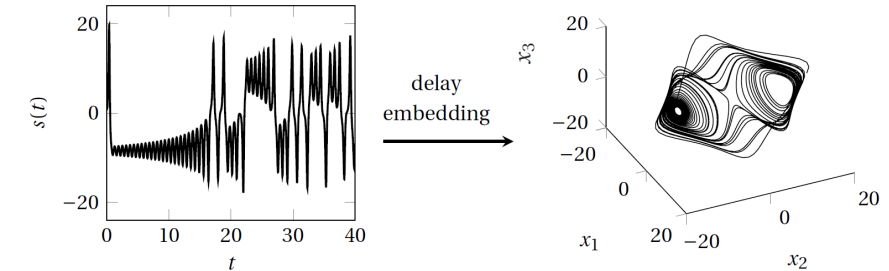
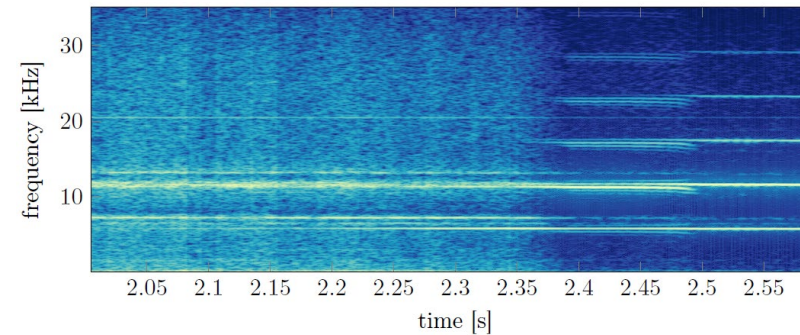
* scarce, largely unavailable

(Vibration) Data is the New Oil

Advanced Time Series Analysis

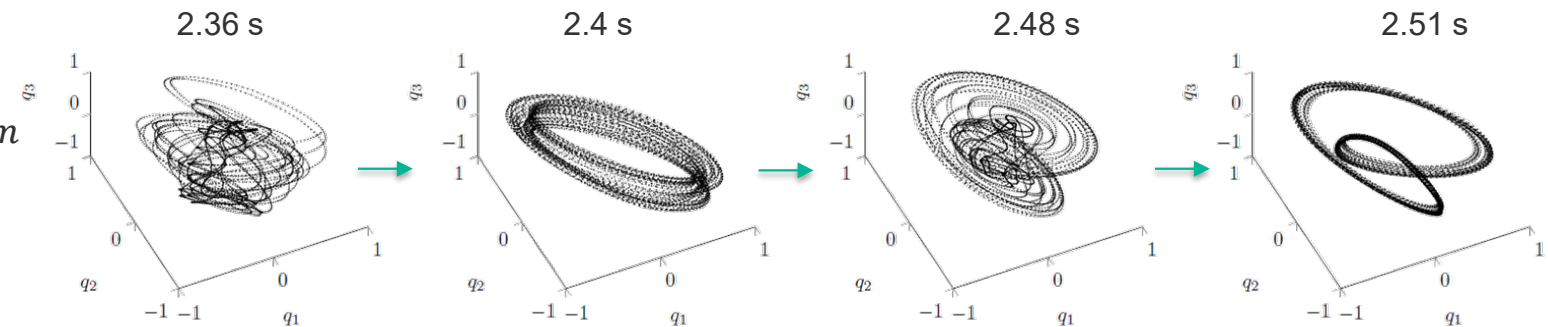


- Classical: FFT



- Delay embedding
(Takens' theorem)

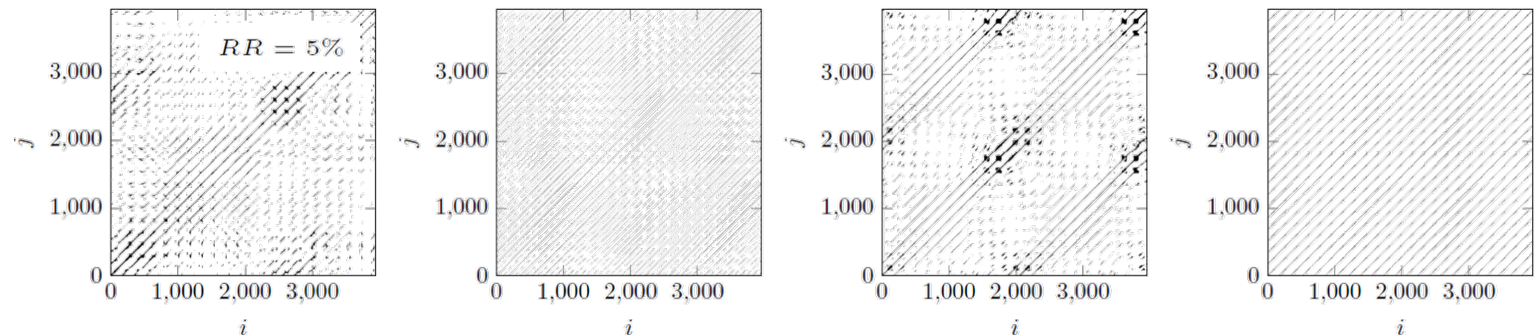
$$x_n \equiv (s_n, \dots, s_{n+(m-1)k}) \in \mathbb{R}^m$$



- Recurrence Plots

$$R_{i,j} = \Theta_{\epsilon}(|x_i - x_j|)$$

feature engineering

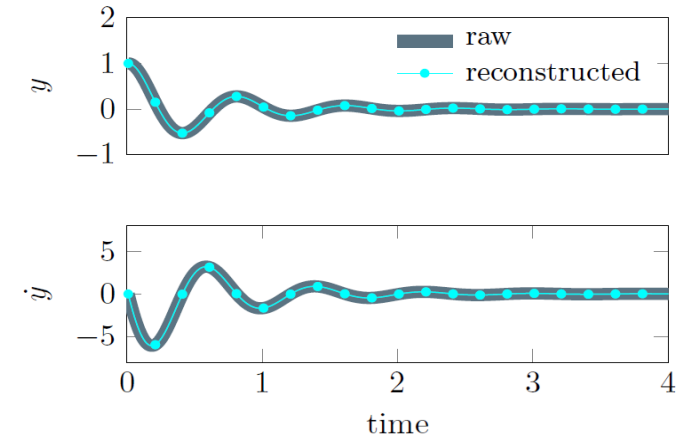
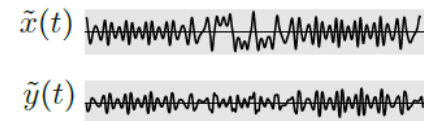
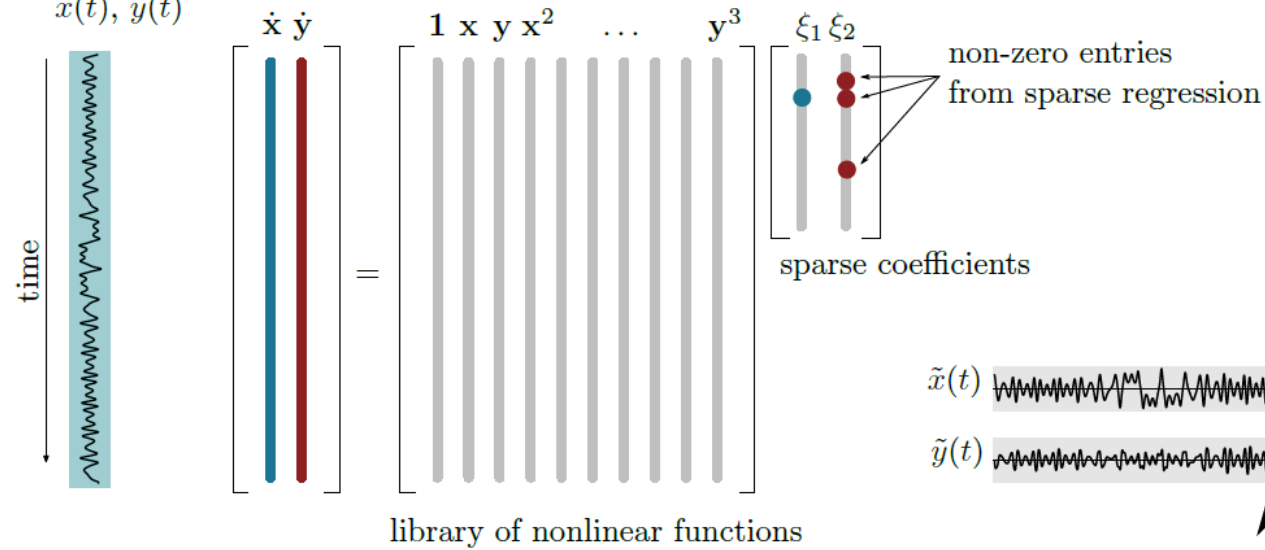


Discovering Equations from Data (SINDy)

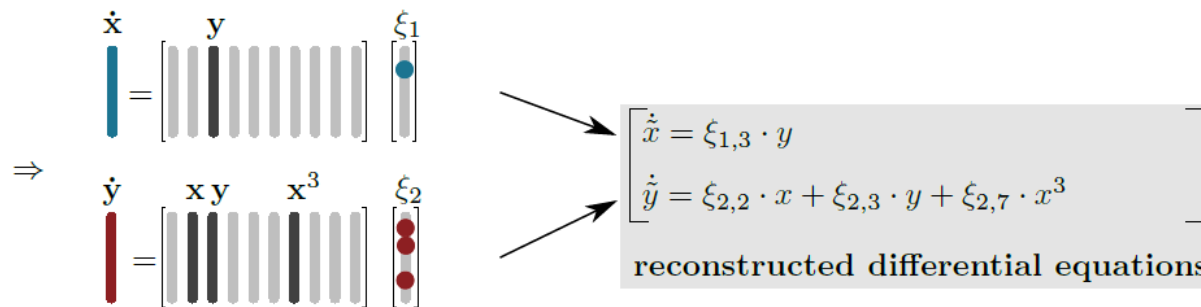


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input: $x(t), y(t)$
 $\dot{x}(t), \dot{y}(t)$



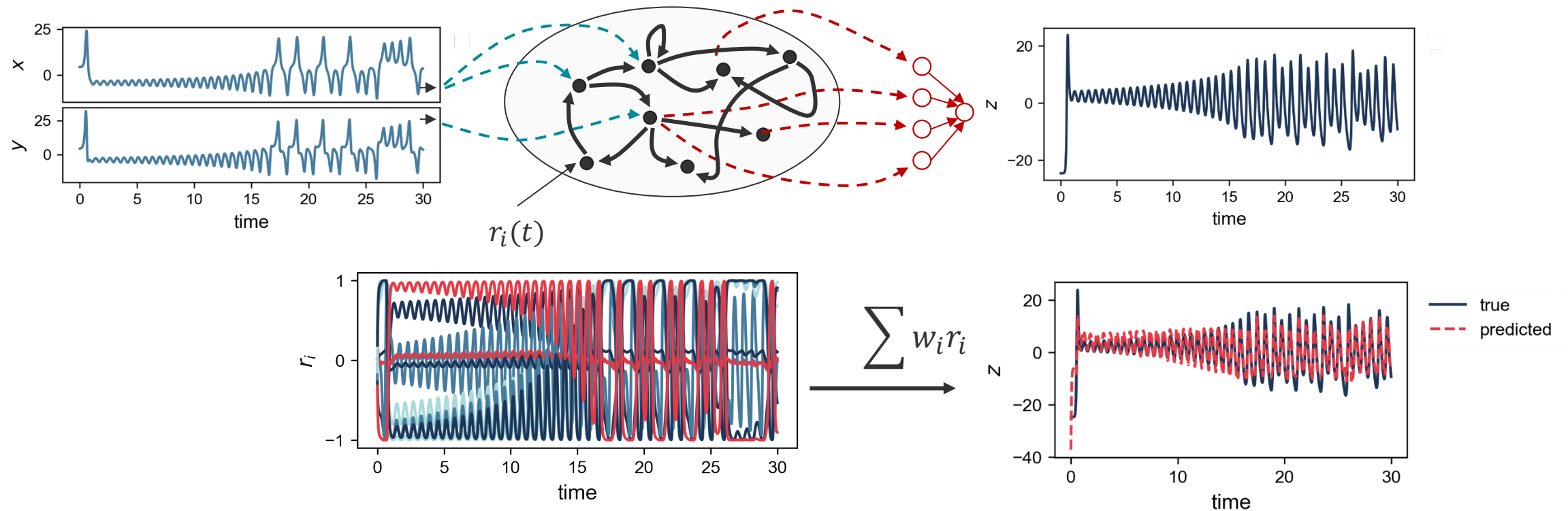
Stender et al. (2019) *Vibration*



Illustrative Example



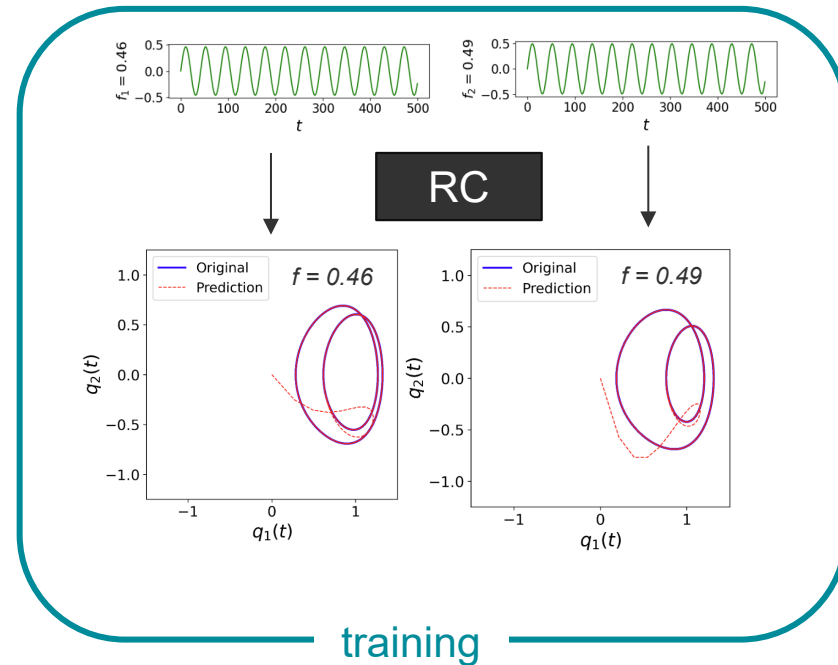
Reconstruction of an unobserved system state (Lorentz system)



Learning Bifurcating Dynamics



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Predicting multi-parametric dynamics of an externally forced oscillator using reservoir computing and minimal data

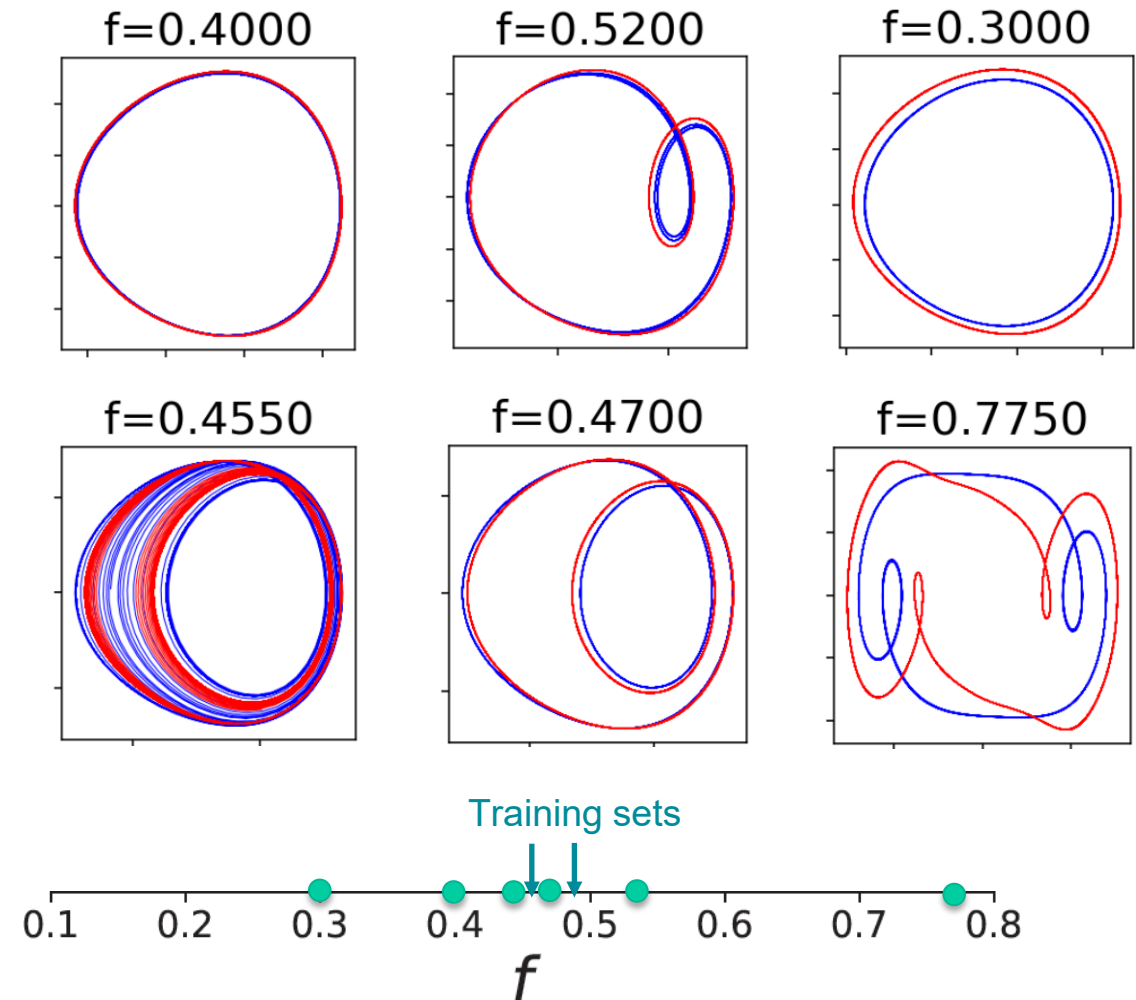
Research | Open access | Published: 05 December 2024
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Nonlinear Dynamics

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Questions?